Statement of Significance

Turbulent premixed flames burn faster than laminar ones through two mechanisms identified by Damköhler: wrinkling by large eddies and enhanced transport by smaller ones. These are typically modelled separately. We show that integrating a spectral eddy diffusivity across the turbulence spectrum yields a closed-form solution for burning velocity, providing deep insight into multiscale turbulence-flame interactions. Our continuous spectral framework shares its form with an established power-law wrinkling model and agrees with measurements across four decades of scale separation. Furthermore, it resolves the consumption speed and flame surface area discrepancy in recent intense turbulence experiments, delivering a parameterisation directly testable against simulations and experiments.